Islam Kathat

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current resume here

islamkhan.in
faiz.14a@gmail.com | +91 80008 02709 | LinkedIn | GitHub

EDUCATION

JECRC UNIVERSITY
MCA in Computer Science
July 2025 | Jaipur, Rajasthan
CGPA: 8.57

MDS UNIVERSITY
BCA in Computer Science
July 2023 | Ajmer, Rajasthan
CGPA: 7.68

SKILLS

PROGRAMMING LANGUAGES
- Java
- SQL
- Python
- JavaScript, TypeScript

ML LIBRARIES
- Scikit-learn
- TensorFlow
- PyTorch (primary)
- NumPy, Pandas, SciPy
- Transformers, PEFT, TRL
- BitsAndBytes, Accelerate

LLM & AI ENGINEERING
- Weights & Biases
- HuggingFace Hub
- Cosine LR, AdamW
- KV Cache, GQA, MoE
- Gradient Checkpointing
- NF4 4-bit, BitsAndBytes
- Mixed Precision (bf16/fp16)
- QLoRA, LoRA, PEFT, SFTTrainer

FULL-STACK ENGINEERING
- Realm, Kafka
- GitHub, GitLab
- React.js, Next.js
- Docker, Firebase, Expo
- React Native, Native Modules
- PostgreSQL, MongoDB, Redis
- Node.js, FastAPI, Flask

SOFT SKILLS
- Rapid Learning
- Public Speaking
- Problem-Solving
- Cross-Team Collaboration
- Technical Documentation

CERTIFICATIONS

Responsible Generative AI (Microsoft)
Deep Learning and AI/ML (KPMG)
Java Certificate (HackerRank)

PROJECTS

LLM PRETRAINING & ALIGNMENT 350M Parameter GPT | GitHub
CUDA | PyTorch | Transformers | Self Attention | Mixed Precision | Hugging Face

- Designed a GPT-style decoder-only transformer (24L × 16H × 1024D, 50304 vocab, RMSNorm, Flash Attention, weight tying) trained on 6.83B tokens, val loss 3.04, perplexity 20.9, competitive with GPT-2 XL at 6× fewer training tokens.
- Solved persistent torch.compile + BF16 OOM issues by isolating `lm_head` and `cross_entropy` outside the compiled graph using `torch.compiler.disable`, preventing AOT Autograd from allocating 12GB FP32 gradient buffers.
- Applied Supervised Fine-Tuning (SFT) on OpenHermes 2.5 (288M tokens, 746K examples), performed identity-alignment tuning with protection examples to prevent factual bleeding; used proper checkpointing logic for resumable training.
- Built a fault-tolerant streaming data pipeline: Dataset streamed + tokenized on a free Colab CPU, saved as `uint16` binary shards, transferred to RunPod via rclone, Monitored training with [Weights & Biases]; published model on [Hugging Face].

MISTRAL-7B FINE-TUNING Fine-tuned an LLM Model | GitHub
QLoRA | PEFT | TRL | PyTorch | BitsAndBytes | Hugging Face

- Fine-tuned Mistral-7B-Instruct-v0.3 using QLoRA (NF4 4-bit, r=64) on 500 domain-specific instruction pairs with 0.5% of trainable parameters.
- Attained 13% perplexity reduction vs base model baseline using cosine LR, Paged AdamW, and gradient checkpointing on Nvidia A100 80GB GPU.
- Logged training metrics via [Weights & Biases], merged the LoRA adapter into the base model, open-sourced the final model on [Hugging Face] with a model card.

RAG SYSTEM Retrieval-Augmented Generation System | Live
FastAPI | ChromaDB | Ollama | SearXNG | Sentence Transformers

- Architected full-stack RAG application using FastAPI, React.js, ChromaDB, and LLaMA 3.1 with Ollama for local LLM inference and Grok API for production.
- Integrated SearXNG and Trafilatura in local and tavily in production with multi-format document ingestion for reducing RAG retrieval time by 17%.
- Achieved sub-600ms retrieval via Sentence Transformer embeddings, 512-char chunking with 100 overlapping chunks and cross-encoder re-ranking.

ADDITIONAL PROJECTS CNN | Mobile | Full Stack | Productivity

- **MNIST Digit Classifier**: Trained MNIST digit classifiers (MLP, SelectKBest, CNN with BatchNorm) from scratch, each achieving 90%+ accuracy in prediction.
- **CIFAR-10 Object Classifier**: Developed and benchmarked CNN (92%), ResNet (95%), and EfficientNetV2B0 transfer learning (97%) classification models.
- **AI Club**: Multi-AI desktop application enabling parallel LLM inference and real-time response comparison from a single prompt saving 15% of time.

EXPERIENCE

JSB GLOBAL INFOTECH | MERN Stack Developer | Certificate
May 2025 - May 2026 | Full Time

- Engineered React Native (Expo) applications and deployed React.js/Next.js web platforms, including admin panels and public-facing frontends.
- Implemented backend microservices using Redis, Docker, AWS S3, and AI pipelines to automate scraping and rewriting automotive blogs/news content.
- Shipped web and mobile apps using REST APIs, RealmDB, and advanced MongoDB pipelines, reducing API response time by 35%.

CODEALPHA | Data Science Intern | Certificate
July 2024 - August 2024 | Remote

- Applied advanced feature engineering and selection techniques on real-world datasets, boosting ML model accuracy by 70% over the unengineered baseline.
- Enhanced model accuracy by 15% through iterative data visualization, hyperparameter tuning, and pipeline optimization across classification tasks.
- Obtained 90% precision on assigned tasks using Scikit-learn and Pandas, consistently delivering all project milestones ahead of deadlines.